THE UNIVERSITY OF AZAD JAMMU AND KASHMIR MUZAFFARABAD



Department of Software Engineering

Subject: Software Testing Lab

Lab Instructor: Engr. Fahad Nisar

Semester: 5th SET

Session: 2024 – 28

Manual Test Execution & Defect Reporting

Lab Task:3

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Roll No: 2024 – SET – 19

1. Objective

The objective of this lab is to execute the ten test cases designed in Lab Task 02 against the same TodoMatic application, record the actual results and execution status, capture evidence, and report the defects observed during testing.

2. Application and Test Environment

Item	Details
Application	TodoMatic (MDN Todo React)
GitHub repository	https://github.com/mdn/todo-react
Application/package version	0.3.0 (package.json version)
Source commit	Not available — project was downloaded as a ZIP archive without Git metadata.
Runtime / local URL	Node.js v24.21.0; Yarn 1.22.22; http://localhost:3000
Operating system	Windows (exact edition/build not recorded)
Browser	Google Chrome (exact version not recorded)
Execution type	Manual testing

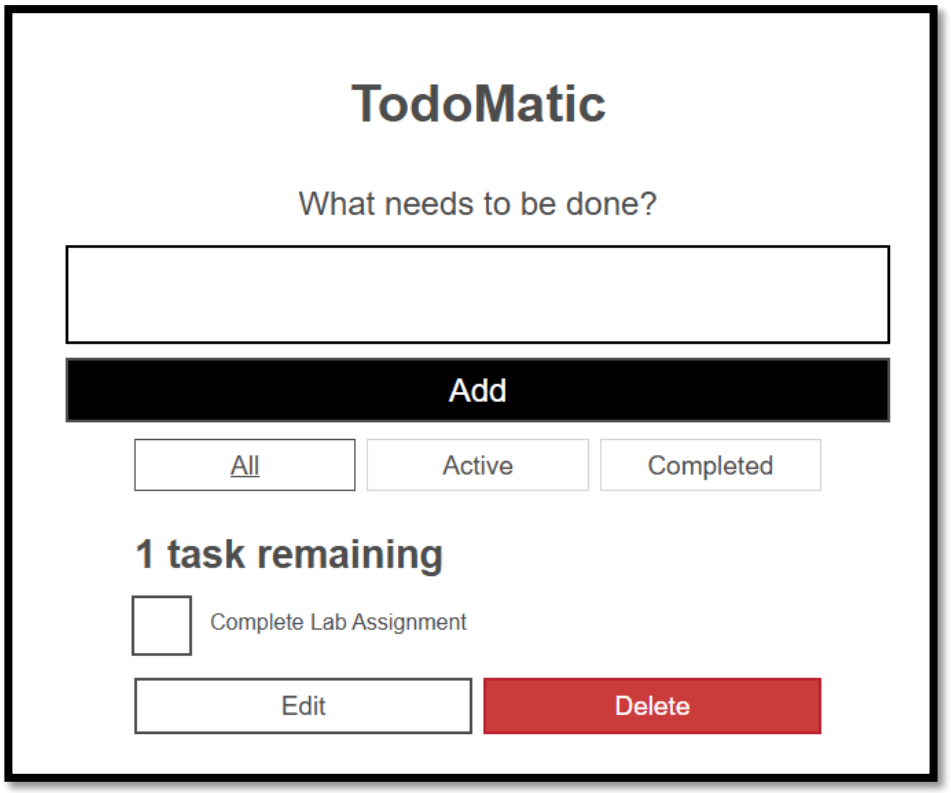
Environment note: The application was run locally from the project directory F:\todo-react-main\todo-react-main using the Vite development command `yarn dev`. The report records only environment details that were available during the test session.

3. Test Execution Results

The following test cases are the same ten cases designed in Lab Task 02. The expected results were kept unchanged. Each case was executed manually in TodoMatic.

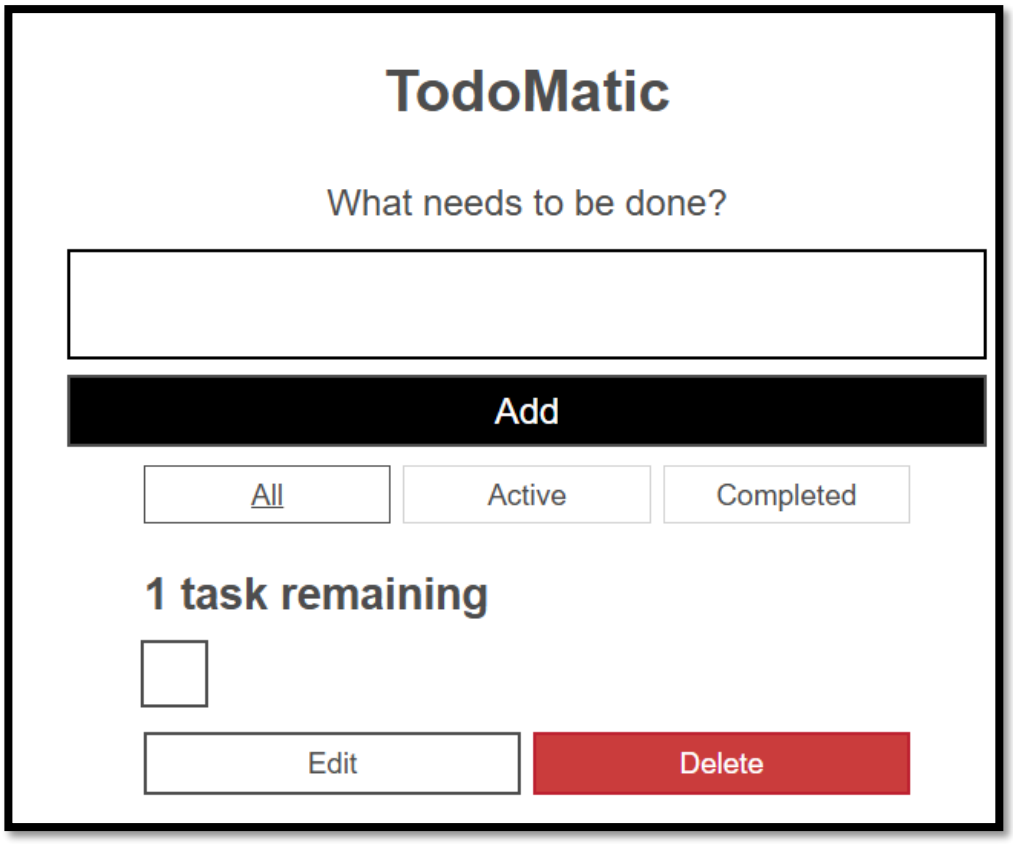
TC-01 — Add Valid Task

Execution field	Recorded details
Requirement / Rule	R-01 — User can add a new task when valid, non-empty text is entered.
Preconditions	TodoMatic is open and the task list is empty.
Test data	Complete Lab Assignment
Steps	1. Enter `Complete Lab Assignment` in the task input field. 2. Click Add. 3. Check the task list and input field.
Expected result	The task appears exactly once in the list and the input field is cleared.
Actual result	`Complete Lab Assignment` was successfully added to the task list.
Status	PASS
Evidence reference	TC-01.png
Remarks	—



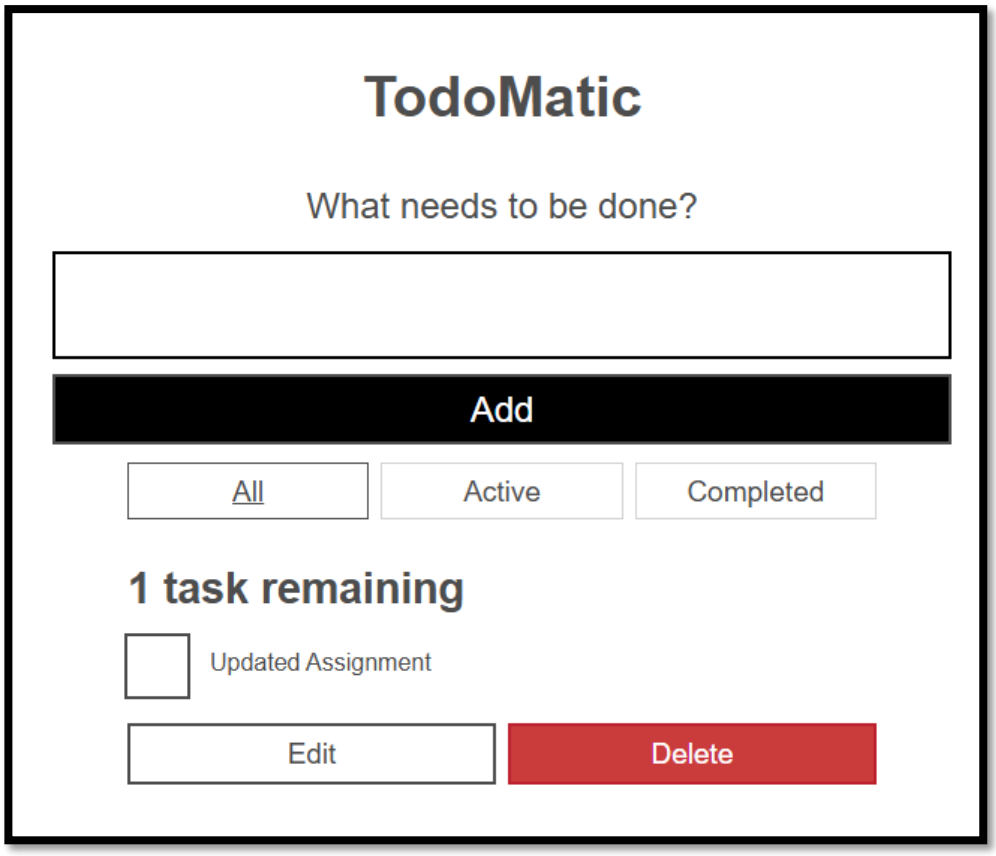
TC-02 — Attempt to Add Empty Task

Execution field	Recorded details
Requirement / Rule	R-02 — The application should not add a task when the input is empty.
Preconditions	TodoMatic is open and the task list is empty.
Test data	Empty input (no characters entered).
Steps	1. Leave the task input field empty. 2. Click Add. 3. Inspect the task list and remaining-task counter.
Expected result	No task is added and the remaining-task count stays at 0.
Actual result	An empty task was added to the list and the remaining-task counter displayed 1.
Status	FAIL
Evidence reference	TC-02.png
Remarks	Linked defect: D-01.



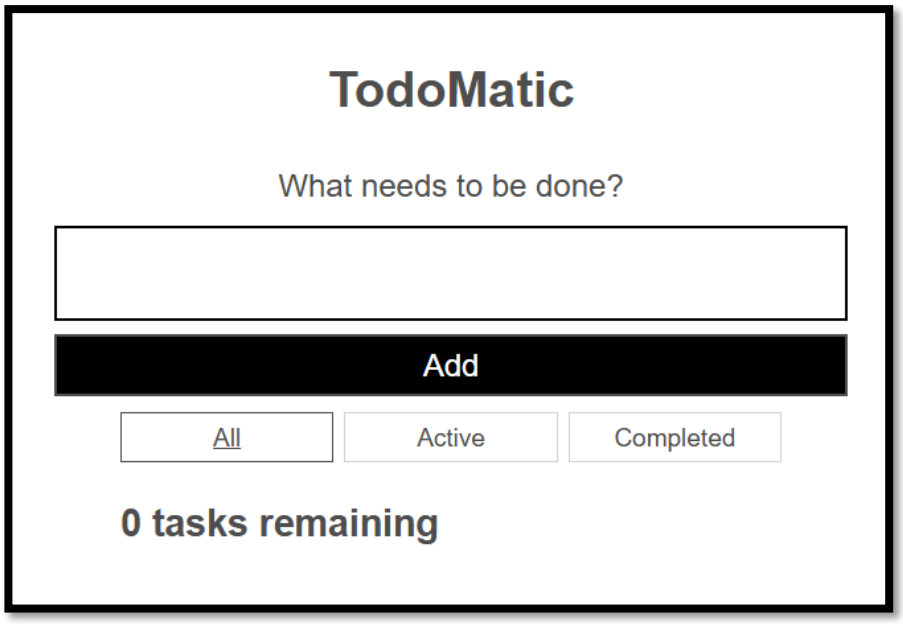
TC-03 — Edit Existing Task

Execution field	Recorded details
Requirement / Rule	R-03 — User can edit the text of an existing task.
Preconditions	A task named `Old Assignment` exists in the list.
Test data	Existing: Old Assignment; replacement: Updated Assignment.
Steps	1. Add `Old Assignment`. 2. Click Edit for that task. 3. Replace the text with `Updated Assignment`. 4. Save/finish editing.
Expected result	The task displays `Updated Assignment`; the old text is no longer displayed.
Actual result	The task text was successfully changed from `Old Assignment` to `Updated Assignment`.
Status	PASS
Evidence reference	TC-03.png
Remarks	—



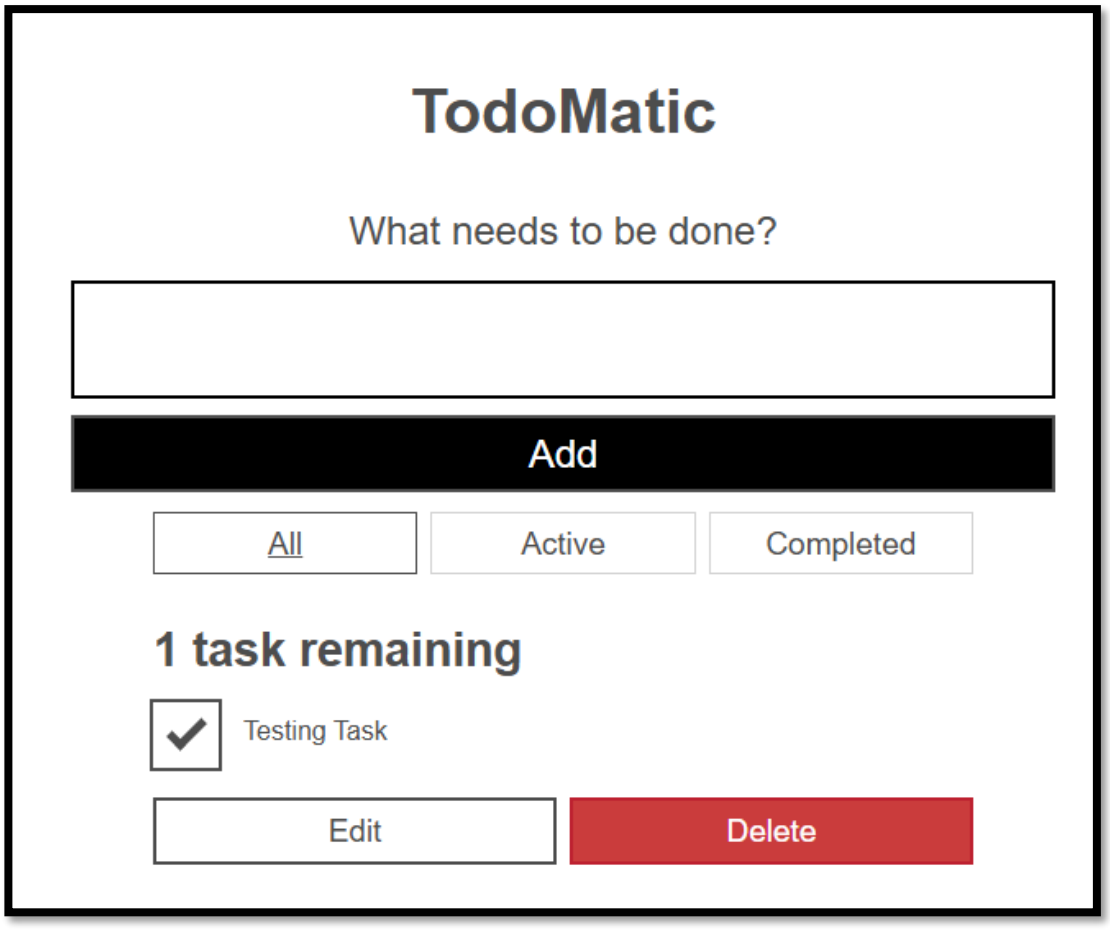
TC-04 — Delete Existing Task

Execution field	Recorded details
Requirement / Rule	R-04 — User can delete an existing task.
Preconditions	A task named `Updated Assignment` exists in the list.
Test data	Updated Assignment
Steps	1. Locate `Updated Assignment`. 2. Click its Delete button. 3. Inspect the task list and counter.
Expected result	The task is removed from the list.
Actual result	`Updated Assignment` was successfully deleted and the task list became empty.
Status	PASS
Evidence reference	TC-04.png
Remarks	—



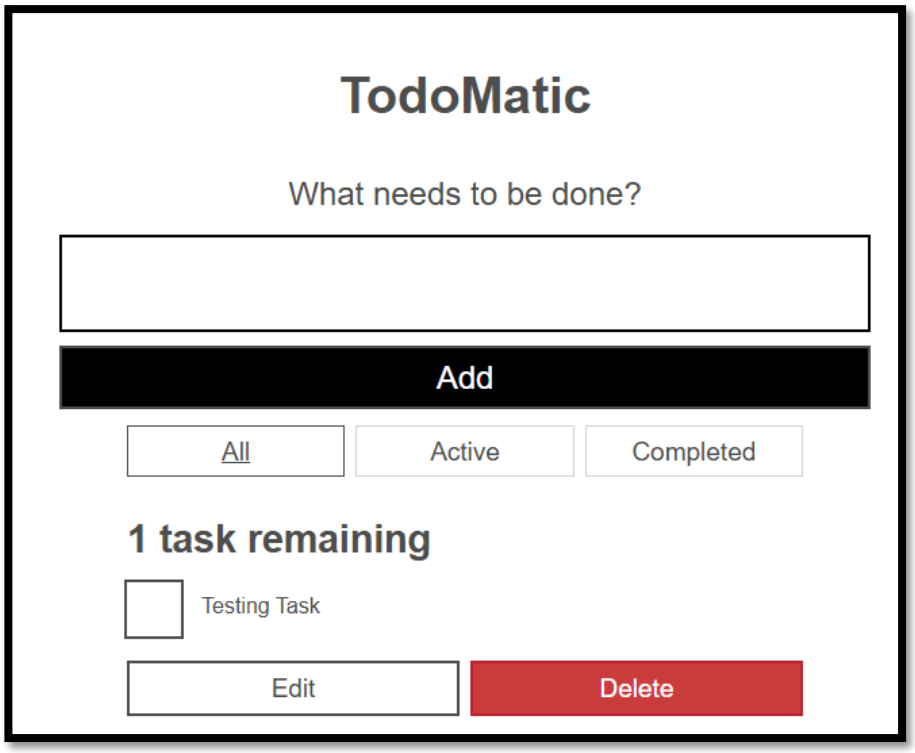
TC-05 — Mark Active Task as Completed

Execution field	Recorded details
Requirement / Rule	R-05 — User can mark an active task as completed.
Preconditions	An active task named `Testing Task` exists.
Test data	Testing Task
Steps	1. Add `Testing Task`. 2. Click its checkbox. 3. Inspect its completion state and remaining-task counter.
Expected result	The task is marked completed and the remaining-task counter becomes 0.
Actual result	The task checkbox showed the completed state, but the remaining-task counter still displayed 1.
Status	FAIL
Evidence reference	TC-05.png
Remarks	Linked defect: D-02 (remaining-task counter does not update correctly).



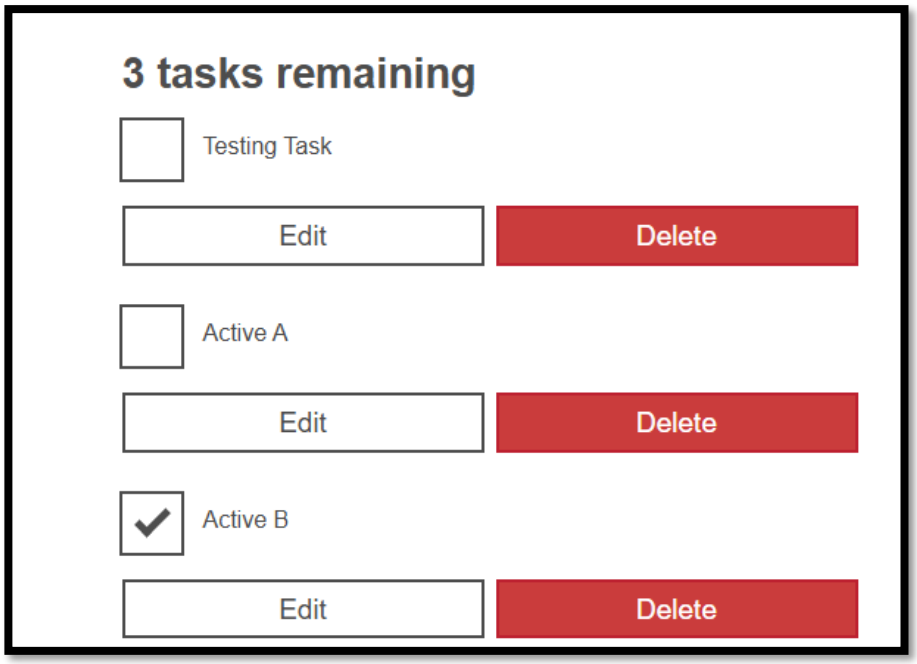
TC-06 — Reactivate Completed Task

2	**Recorded details**
Requirement / Rule	R-06 — User can mark a completed task active again.
Preconditions	A completed task named `Testing Task` exists.
Test data	Testing Task
Steps	1. Locate the completed `Testing Task`. 2. Click its checked checkbox again. 3. Inspect its state and remaining-task counter.
Expected result	The task becomes active again and is counted as one remaining task.
Actual result	The task became active again and the remaining-task counter displayed 1.
Status	PASS
Evidence reference	TC-06.png
Remarks	—



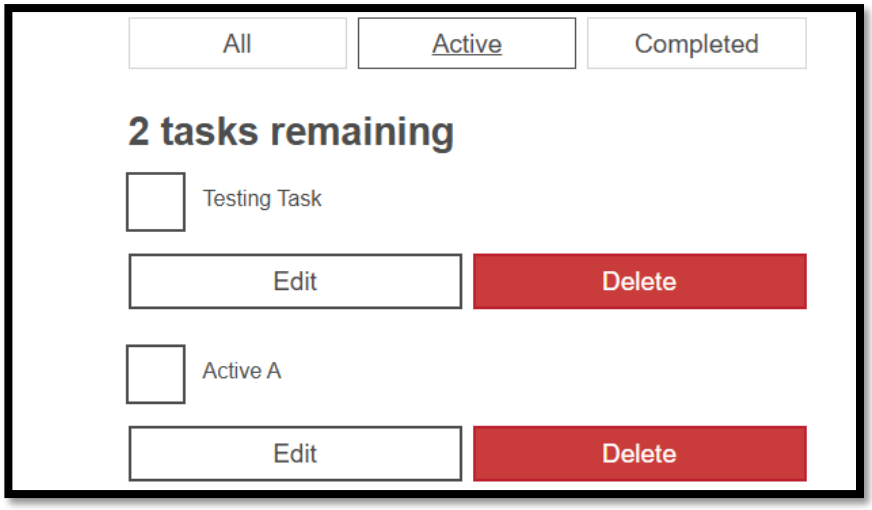
TC-07 — All Filter

Execution field	Recorded details
Requirement / Rule	R-07 — All filter displays both active and completed tasks.
Preconditions	Three tasks exist: two active and one completed.
Test data	Testing Task (active); Active A (active); Active B (completed).
Steps	1. Ensure the three specified tasks exist with the stated completion states. 2. Click All. 3. Inspect the displayed tasks.
Expected result	All three tasks (two active and one completed) are displayed.
Actual result	The All filter displayed all three tasks: two active and one completed.
Status	PASS
Evidence reference	TC-07.png
Remarks	—



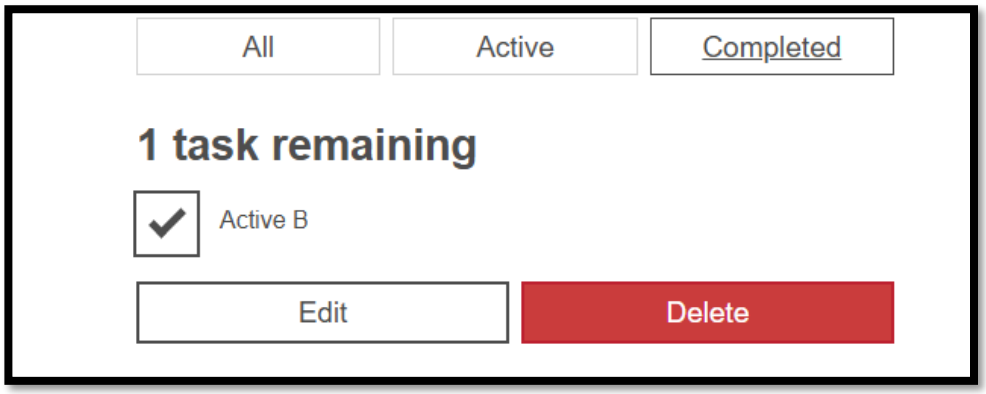
TC-08 — Active Filter

Execution field	Recorded details
Requirement / Rule	R-08 — Active filter displays only incomplete tasks.
Preconditions	Two active tasks and one completed task exist.
Test data	Testing Task (active); Active A (active); Active B (completed).
Steps	1. Ensure the three specified tasks exist with the stated completion states. 2. Click Active. 3. Inspect the displayed tasks.
Expected result	Only the two active tasks are displayed; the completed task is hidden.
Actual result	The Active filter displayed two active tasks and did not display the completed task.
Status	PASS
Evidence reference	TC-08.png
Remarks	—



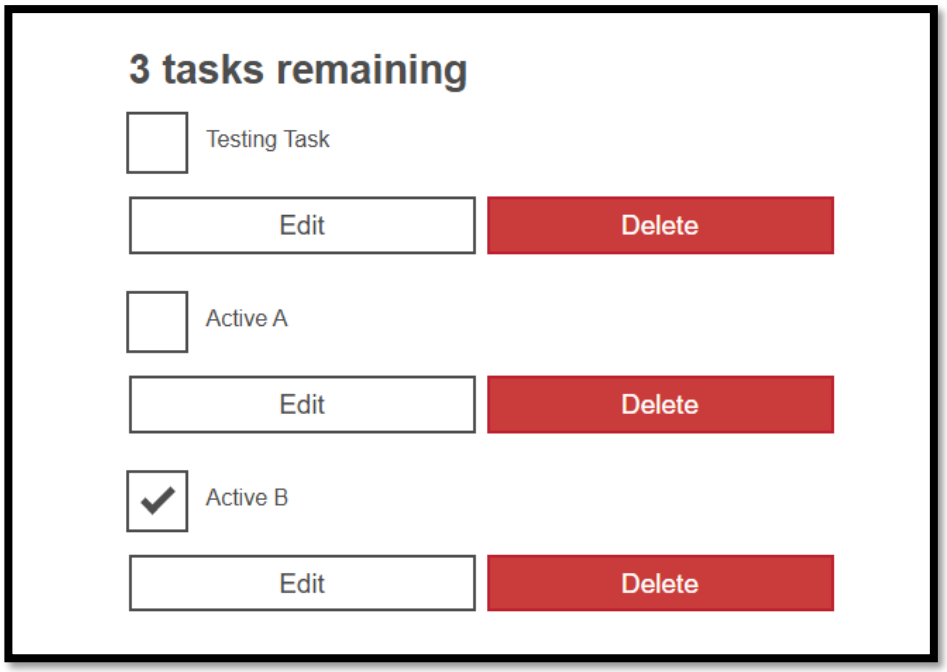
TC-09 — Completed Filter

Execution field	Recorded details
Requirement / Rule	R-09 — Completed filter displays only completed tasks.
Preconditions	Two active tasks and one completed task exist.
Test data	Testing Task (active); Active A (active); Active B (completed).
Steps	1. Ensure the three specified tasks exist with the stated completion states. 2. Click Completed. 3. Inspect the displayed tasks.
Expected result	Only the completed task is displayed; active tasks are hidden.
Actual result	The Completed filter displayed only the completed task.
Status	PASS
Evidence reference	TC-09.png
Remarks	—



TC-10 — Remaining Task Counter

Execution field	Recorded details
Requirement / Rule	R-10 — Remaining-task counter displays the number of active/incomplete tasks.
Preconditions	Three tasks exist: two active and one completed.
Test data	Testing Task (active); Active A (active); Active B (completed).
Steps	1. Ensure the three specified tasks exist with the stated completion states. 2. Click All. 3. Check the remaining-task counter.
Expected result	The counter displays 2 tasks remaining.
Actual result	The counter displayed 3 remaining, although only two tasks were active and one was completed.
Status	FAIL
Evidence reference	TC-10.png
Remarks	Linked defect: D-02.



4. Defect Reports

Two distinct defects were observed. The counter mismatch seen in TC-05 and TC-10 is recorded as one defect because both test cases show the same incorrect remaining-task count behaviour.

D-01 — Empty Task Can Be Added

Defect field	Details
Related test case	TC-02
Version / environment	TodoMatic package version 0.3.0; Windows; Google Chrome; local run at http://localhost:3000.
Preconditions	Application is open and task list is empty.
Steps to reproduce	1. Leave the task input field empty. 2. Click Add. 3. Inspect the task list and remaining counter.
Expected result	The application should reject the empty input; no task should be added and the count should remain 0.
Actual result	An empty task was added and the counter displayed 1.
Evidence	TC-02.png (insert screenshot below).

D-02 — Remaining-Task Counter Counts Incorrectly

Defect field	Details
Related test cases	TC-05 and TC-10
Version / environment	TodoMatic package version 0.3.0; Windows; Google Chrome; local run at http://localhost:3000.
Preconditions	For TC-05: one task is marked completed. For TC-10: two tasks are active and one task is completed.
Steps to reproduce	TC-05: Add `Testing Task`, mark it completed, and inspect the counter. TC-10: Set up two active tasks and one completed task, select All, and inspect

	the counter.
Expected result	TC-05: counter should become 0. TC-10: counter should display 2.
Actual result	TC-05: counter remained 1 after completing the task. TC-10: counter displayed 3 although only two tasks were active.
Evidence	TC-05.png and TC-10.png (insert screenshots below).

5. Execution Summary

Summary item	Count / details
Total test cases	10
Pass	7 — TC-01, TC-03, TC-04, TC-06, TC-07, TC-08, TC-09
Fail	3 — TC-02, TC-05, TC-10
Blocked	0
Not run	0
Observed distinct defects	2 — D-01 and D-02
Remaining work	Insert the saved screenshots into the evidence placeholders; review the final document before submission.

Result interpretation: TC-05 and TC-10 both expose the same remaining-task counter problem, so they are linked to one distinct defect (D-02), not counted as two separate defects.

6. Conclusion

All ten Lab Task 02 test cases were manually executed against TodoMatic. Seven test cases passed and three failed. Two distinct defects were observed: the application accepts an empty task, and the remaining-task counter reports an incorrect value in the tested scenarios. No test cases were blocked or left unrun. The screenshots listed in the evidence references should be inserted in the reserved spaces before submission.